

1. Business Problem:

Superstore generates approximately **\$2.30M** in sales but only **\$286.40K** in profit, resulting in a relatively low **12.47% profit margin**. Despite strong revenue, certain product categories contribute disproportionately to low profitability. The objective of this analysis is to identify the underlying causes of poor profitability, particularly within the Furniture category, and provide data-driven recommendations to improve business performance.

2. Business Objectives:

This investigation aims to identify the primary factors responsible for poor profitability and recommend actionable improvements.

- Analyse the overall sales and profitability of the Superstore.
- Identify the best and worst performing product categories and sub-categories
- Investigate factors contributing to losses in the Furniture category.
- Evaluate the impact of discounts, regions, customer segments and individual products on profitability.
- Provide data-driven recommendations to improve overall business profitability.

3. Business Questions:

Overall Performance

- What are the total sales, total profit, total quantity sold, and profit margin?
- Does the current profit justify the overall sales performance?

Category Analysis

- Which product category generates the highest sales?
- Which product category generates the highest profit?
- Why does the Furniture category generate high sales but comparatively low profit?

Sub-Category Analysis

- Which sub-categories contribute the highest profits?
- Which sub-categories are responsible for the largest losses?

Discount Analysis

- Do higher discounts negatively impact profitability?
- Which sub-categories receive the highest average discounts?

Regional Analysis

- Are losses concentrated in specific regions or spread across all regions?

Customer Analysis

- Which customer segment purchases the most loss-making products?

Product Analysis

- Which products contribute the most to overall losses?

Business Strategy

- What strategies can improve profitability while maintaining strong sales?

4. Dataset Description

- **Dataset Name:** Sample Superstore
- **Source:** Tableau Sample Superstore Dataset
- **Number of Records:** 9,994
- **Number of Columns:** 21
- **Time Period:** 2014–2017
- **Granularity:** Each row represents a unique product purchased within an order.
- **Key Variables:**
 - Order ID
 - Order Date
 - Customer ID
 - Product Name
 - Category
 - Sub-Category
 - Region
 - Sales
 - Quantity
 - Discount
 - Profit

5. Data Dictionary:

Column	Description
Order ID	Unique identifier for each order
Customer ID	Unique identifier for each customer
Product ID	Unique identifier for each product
Sales	Revenue generated from the order
Profit	Profit earned from the order

Column	Description
Discount	Discount applied to the order
Quantity	Number of units purchased
Category	Product category
Sub-Category	Product sub-category
Region	Sales region

6. Data Cleaning & Quality Assessment:

6.1 Data Import

The dataset was initially explored in Microsoft Excel to understand its structure and validate data quality. It subsequently imported into MySQL for exploratory data analysis and finally into Power BI for interactive dashboard development.

6.2 Data Quality Checks:

The dataset was assessed for common data quality issues before analysis, including:

- Missing Values
- Duplicate records
- Incorrect data types
- Invalid values
- Import inconsistencies

6.3 Dataset Summary:

Metric	Value
Total Rows	9994
Total Columns	21
Missing Values	0
Duplicate Rows	0
Invalid Sales	0
Invalid Quantity	0

The dataset consists of 9994 records and 21 columns. No missing values or duplicate records were identified, indicating good overall data quality.

6.4 Missing Value Analysis:

Missing values were checked across all columns using Excel and SQL. No missing values were found in any columns, indicating that dataset was complete and suitable for analysis.

6.5 Duplicate Analysis:

Duplicate records were checked using Row ID. All Row IDs were unique, confirming that no duplicate records existed in the dataset.

6.6 Data Type Validation:

Datatypes were validated before analysis. During the Power BI import process, the Order Date and Ship Date columns generated parsing errors due to inconsistent date formats within the source CSV. The issue was resolved by importing the dates using the appropriate locale settings, ensuring accurate date conversion and subsequent analysis.

6.7 Import Validation:

During the initial MySQL import, only 9694 records were imported because some rows were rejected due to date format inconsistencies. After investigating the issue, the dataset was re-imported correctly, resulting in the successful import of all 9994 records.

6.8 Initial Observations:

- The dataset contains three product categories and multiple sub-categories.
- Sales, Profit, Discount and Quantity are the primary numerical variables.
- Customer, Product, Region, and Category fields provide multiple dimensions for analysis.
- The dataset spans four years (2014-2017), enabling trend analysis.

6.9 Data Cleaning Summary:

Overall, the dataset required minimal cleaning. The primary challenge involved restoring inconsistent date formats during import. Once corrected, the dataset was verified to contain complete, unique, and analysis-ready records.

7. Exploratory Data Analysis (EDA):

7.1 Overall Business Performance

Business Question: How is business performing overall?

SQL Query:

```
SELECT  
    round(SUM(sales),2) TOTAL_SALES,  
    ROUND(SUM(Profit),2) TOTAL_PROFIT,
```

```

SUM(Quantity) TOTAL_QTY,
COUNT(DISTINCT `Order ID`) TOTAL_ORDERS
FROM superstore;

```

Result:

Metric	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Quantity	37,873
Total Orders	5,009

Observation:

- The Superstore generated approximately \$2.30M in sales
- Overall profit was \$286.40K.
- Approximately 37.9K products were sold across 5009 unique orders.

Business Insight:

Although the business generates substantial revenue, profitability is relatively modest compared to total sales. This indicates that high sales alone do not necessarily translate into strong financial performance, warranting further investigation into underlying causes.

7.2 Category Analysis

Business Question: Which product categories generate the highest sales and profit?

SQL Query:

```

SELECT Category, ROUND(SUM(Sales),2) TOTAL_SALES, ROUND(SUM(Profit),2) TOTAL_PROFIT
FROM superstore
GROUP BY 1
ORDER BY 3 DESC;

```

Result:

Category	TOTAL_SALES	TOTAL_PROFIT
Technology	836154.03	145454.95
Office Supplies	719047.03	122490.80

Category	TOTAL_SALES	TOTAL_PROFIT
Furniture	741999.80	18451.27

Observation:

- Technology is the highest-performing category in terms of both sales and profit.
- Furniture generates the second highest sales but contributes the lowest profit.

Business Insight:

The Furniture category demonstrates an imbalance between revenue generation and profitability, making it the primary area for further investigation.

7.3 Sub-Category Analysis

Business Question: Which Furniture sub-categories contribute to the losses?

SQL Query:

```
SELECT `Sub-Category`, ROUND(SUM(Sales),2) TOTAL_SALES, ROUND(SUM(Profit),2)
TOTAL_PROFIT
FROM superstore
GROUP BY `Sub-Category`
ORDER BY 3;
```

Result:

Tables	206965.53	-17725.48
Bookcases	114880.00	-3472.56
Supplies	46673.54	-1189.10
Fasteners	3024.28	949.52
Machines	189238.63	3384.76
Labels	12486.31	5546.25
Art	27118.79	6527.79
Envelopes	16476.40	6964.18
Furnishings	91705.16	13059.14
Appliances	107532.16	18138.01

Storage	223843.61	21278.83
Chairs	328449.10	26590.17
Binders	203412.73	30221.76
Paper	78479.21	34053.57
Accessories	167380.32	41936.64
Phones	330007.05	44515.73
Copiers	149528.03	55617.82

Observation:

- Tables generate substantial revenue but incur the largest loss
- Bookcases also operate at a loss despite significant sales

Business Insight:

Losses within the Furniture category are concentrated in two sub-categories rather than being spread evenly across all furniture products.

7.4 Discount Analysis

Business Question: Do higher discounts contribute to lower profitability within Furniture category.

SQL Query:

```
SELECT
  `Sub-Category`,
  AVG(Discount)*100 AVG_DISCOUNT_PERC,
  ROUND(SUM(Sales),2) TOTAL_SALES,
  ROUND(SUM(Profit),2) TOTAL_PROFIT
FROM superstore
WHERE `Sub-Category` IN ('Tables','Bookcases')
GROUP BY 1
ORDER BY 2 DESC;
```

Result:

Tables	26.128527	206965.53	-17725.48
Bookcases	21.114035	114880.00	-3472.56

Observation:

- Tables receive an average discount of approximately 26%, the highest among the two loss-making sub-categories.
- Bookcases receive an average discount of approximately 21%.
- Both sub-categories operate at loss despite considerable sales.

Business Insight:

Higher discounts appear to contribute to reduced profitability. However, discounts alone do not fully explain the losses, as some highly discounted sub-categories remain profitable. Therefore, additional factors must be influencing profitability.

Hypothesis Validation:

- Hypothesis 1: High discounts are the primary reason for losses.
- Conclusion: Partially Supported. High discounts contribute to lower profitability, but they are not the sole cause of the losses.

7.5 Regional Analysis

Business Question: Are losses concentrated in specific regions?

SQL Query:

```
SELECT
  `Sub-Category`,
  Region,
  ROUND(SUM(Sales),2) TOTAL_SALES,
  ROUND(SUM(Profit),2) TOTAL_PROFIT
FROM superstore
WHERE `Sub-Category` IN ('Tables','Bookcases')
GROUP BY 1,2
ORDER BY TOTAL_PROFIT;
```

Result:

Tables	East	39139.81	-11025.38
Tables	South	43916.19	-4623.06
Tables	Central	39154.97	-3559.65

Bookcases	Central	24157.18	-1997.90
Bookcases	West	36004.12	-1646.51
Bookcases	East	43819.33	-1167.63
Bookcases	South	10899.36	1339.49
Tables	West	84754.56	1482.61

Observation:

- Tables generate losses in all three regions except **West** region
- Bookcases generate losses in most regions but remain profitable in **South** region
- The **East** region contributes the largest overall losses.

Business Insight:

The losses are not isolated to a single region. Instead, they occur across multiple regions, indicating that the issue is systemic rather than region-specific.

Hypothesis Validation:

- Hypothesis 2: Losses are concentrated within specific regions.
- Conclusion: Partially Supported. Regional performance influences profitability, but losses are widespread across most regions.

7.6 Product Analysis

Business Question: Are a small number of products responsible for the majority of losses?

SQL Query:

```
SELECT
    `Sub-Category`,
    `Product Name`,
    ROUND(SUM(Sales),2) TOTAL_SALES,
    ROUND(SUM(Profit),2) TOTAL_PROFIT,
    AVG(Discount)*100 AVG_DISCOUNT
FROM superstore
WHERE `Sub-Category` IN ('Tables','Bookcases')
GROUP BY 1,2
```

ORDER BY TOTAL_PROFIT

LIMIT 10;

Result:

Tables	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	9917.6 4	- 2876.1 2	28.00000 0
Tables	Bush Advantage Collection Racetrack Conference Table	9544.7 3	- 1934.4 0	35.00000 0
Tables	Balt Solid Wood Round Tables	6518.7 5	- 1201.0 6	20.00000 0
Tables	BoxOffice By Design Rectangula r and Half- Moon Meeting Room Tables	1706.2 5	- 1148.4 4	48.33333 3
Tables	Riverside Furniture Oval Coffee Table, Oval End Table, End Table with Drawer	4446.1 8	- 1147.4 0	30.00000 0
Bookcase s	O'Sullivan 4-Shelf Bookcase	2740.2 0	-975.10	42.14285 7

	in Odessa Pine			
	Bretford “Just In Time” Height- Adjustable Multi-Task Work Tables	5634.9 0	-964.19	22.50000 0
Tables	Bevis Oval Conference Table, Walnut	6942.0 7	-856.01	27.50000 0
Tables	BPI Conference Tables	2241.8 7	-795.97	37.00000 0
Bookcase s	O'Sullivan Plantations 2-Door Library in Landvery Oak	2946.3 7	-767.74	33.20000 0

Observation:

- Several conference tables and bookcases consistently appear among the highest loss-making products.
- Approximately 60 products contributed to losses within these sub-categories.

Business Insight:

Losses are concentrated among a relatively small group of products rather than across the entire product portfolio.

Hypothesis Validation:

- Hypothesis 3: A few products contribute disproportionately to the overall losses.
- Conclusion: Supported. The analysis confirms that a limited number of products account for a significant share of the losses.

7.7 Customer Segment Analysis

Business Question: Which customer segment purchases the loss-making products most frequently?

SQL Query:

```
SELECT
  `Sub-Category`,
  Segment,
  ROUND(SUM(Sales),2) TOTAL_SALES,
  ROUND(SUM(Profit),2) TOTAL_PROFIT,
  AVG(DISCOUNT)*100 AVG_DISCOUNT
FROM superstore
WHERE `Sub-Category` IN ('Tables','Bookcases')
GROUP BY 1,2
ORDER BY TOTAL_PROFIT;
```

Result:

Tables	Consumer	99933.80	-9728.04	26.603774
Tables	Corporate	70871.72	-4906.50	25.990566
Bookcases	Consumer	68632.73	-4435.64	22.145038
Tables	Home Office	36160.02	-3090.94	25.000000
Bookcases	Home Office	12241.34	324.63	15.545455
Bookcases	Corporate	34005.92	638.45	21.875000

Observation:

- The Consumer segment contributes to the highest sales for both Tables and Bookcases.
- Corporate customers rank second, while Home Office customers contribute the least.

Business Insight:

Since the Customer segment purchases the largest volume of loss-making products, pricing and promotional strategies targeting this segment should be carefully reviewed.

Hypothesis Validation:

- Hypothesis 4: Consumers purchase the majority of the loss-making products.
- Conclusion: Supported. The Consumer segment is the primary purchaser of the loss-making products.

7.8 Root Cause Summary:

The investigation identified multiple factors contributing to the poor profitability of the Furniture category:

- Tables and Bookcases are the only Furniture sub-categories operating at a loss.
- Both receive among the highest average discounts, reducing overall profitability.
- Losses occur across multiple regions, indicating that the issue is not limited to a single market.
- A relatively small number of products account for a significant proportion of the total losses.
- The Consumer segment contributes the highest sales of these loss-making products.

These findings suggest that the issue stems from a combination of pricing strategy, discounting practices, and product-level performance rather than a single isolated factor.

8. Key Findings

The analysis identified several important insights regarding Superstore's sales and profitability:

- The Superstore generated approximately \$2.30M in sales and \$286.40K in profit, resulting in an overall 12.47% profit margin.
- Technology is the highest-performing category in terms of both sales and profitability.

- Furniture generates the second-highest sales (~\$742K) but contributes the lowest profit (~\$18K), indicating poor profitability.
- Within the Furniture category, Tables (-\$17.7K) and Bookcases (-\$3.4K) are the only loss-making sub-categories.
- Tables (26%) and Bookcases (21%) receive among the highest average discounts, which negatively impacts profitability.
- Losses are observed across multiple regions, with the East region contributing the largest losses.
- The Consumer segment purchases the highest volume of the identified loss-making products.
- A relatively small number of products account for a significant proportion of the overall losses, suggesting that targeted interventions would be more effective than broad category-level changes.

9. Business Recommendations:

Based on the analysis, the following recommendations are proposed to improve profitability:

Pricing Strategy

- Review the discount policy for Tables and Bookcases to reduce excessive discounting while maintaining competitiveness.
- Reassess the pricing strategy for consistently loss-making products.

Product Portfolio

- Identify and discontinue or redesign products that consistently generate losses.
- Prioritize the promotion of profitable Furniture products instead of heavily discounting low-performing items.

Regional Strategy

- Conduct a detailed review of pricing and promotional strategies in the East region to identify region-specific challenges.
- Monitor regional profitability regularly before introducing large-scale promotional campaigns.

Customer Strategy

- Evaluate promotional offers targeted at the Consumer segment to ensure discounts do not significantly reduce profitability.
- Develop targeted marketing campaigns that encourage purchases of higher-margin products.

Business Monitoring

- Establish a periodic product profitability review to identify underperforming products early.
- Develop interactive dashboards to continuously monitor sales, discounts, and profitability across categories and regions.

10. Conclusion:

This analysis successfully identified the primary factors affecting the profitability of the Superstore business.

Although the Superstore demonstrates strong sales performance, profitability is uneven across product categories. The investigation revealed that the Furniture category, particularly the Tables and Bookcases sub-categories, is the primary contributor to reduced profitability. High discount levels, widespread regional losses, and a limited number of consistently loss-making products collectively contribute to this issue.

By implementing targeted pricing strategies, optimizing discount policies, reviewing product performance, and continuously monitoring profitability through interactive dashboards, the business can improve overall profitability while maintaining strong sales performance.

11. Tools & Technologies used:

Category	Tools
Data Cleaning	Microsoft Excel
Database	MySQL 8.3
Query Language	SQL
Data Visualization	Power BI
Documentation	Microsoft Word
Version Control	Git & GitHub